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# **Lucy Mission LDD**

**Lucy Archive Team**

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# USER GUIDE

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The *Lucy* mission's local data dictionary is used to describe and enumerate *Lucy* mission specific classes and attributes in PDS4 labels. It also includes Lucy Instrumentation specific classes and attributes for those instruments that require instrument nuanced metadata.

{ August 12, 2024 }

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## INTRODUCTION

1. Purpose of this User's Guide
2. Audience
3. Applicable Documents





## **OVERVIEW OF THE *LUCY* LOCAL DATA DICTIONARY**

The *Lucy* mission's local data dictionary is used to describe and enumerate *Lucy* Mission specific classes and attributes in PDS4 labels. It also includes Lucy Instrumentation specific classes and attributes.



## ORGANIZATION OF CLASSES AND ATTRIBUTES

The *Lucy* mission dictionary is organized as set of four general classes and four instrument specific classes. The general classes are used to describe either observational parameters common to all mission instruments, or general data processing parameters. The instrument specific classes are used to describe instrument specific metadata that does not easily fit into a discipline specific data dictionary due to a mission specific nuance.

The *Lucy* mission dictionary hierarchy is an alphabetical list of the eight main classes. There is not a generic wrapper class. The eight main classes are as follows:

```
LEISA_Instrument_Parameters
LLORRI_Instrument_Parameters
LRalph_Instrument_Common_Parameters
Lucy_Observation_Planning
Lucy_Observation_Time_Information
Lucy_Product_Information
Lucy_Target_List
MVIC_Instrument_Parameters
```

General classes used for all Lucy products begin with the “Lucy\_”, whereas instrument specific classes begin with the instrument or instrument suite name. Instrument specific classes may contain sub-classes. Use case examples for each class can be found in the Examples section of this document.

### 3.1 Lucy\_Observation\_Time\_Information

The *Lucy\_Observation\_Time\_Information* class is used to record the observation identification and timing information attached to each science observation sequence. The attributes in this class are listed below:

```
lucy:observation_id
lucy:observation_id_count
lucy:observation_complete
lucy:observation_missing_packets
lucy:start_sclk
lucy:mid_sclk
lucy:stop_sclk
lucy:mid_utc
lucy:mid_sclk_string
lucy:mid_utc_doy
lucy:mid_utc_jd
lucy:mid_ephemeris_time
lucy:ccsds_sclk_time
```

The `observation_id` attribute is the only required element.

### 3.2 Lucy\_Observation\_Planning

The `Lucy_Observation_Planning` class is a general *Lucy* class used to record the science planning metadata that describes how the observation sequence was planned and the status of the instruments during that observation. The attributes included in this class are as follows:

```
lucy:load_identifier
lucy:mission_segment
lucy:sap_identifier
lucy:visit_name
lucy:instrument_side
lucy:llorri_status
lucy:lralph_status
lucy:ltes_status
lucy:tccam_status
```

The `mission_segment` attribute is the only required element in this class.

### 3.3 Lucy\_Target\_List

The `Lucy_Target_List` class and attributes describe the list of potential targets in a given science observation. This information is supplemental to the `Target_Identification` class found in all observational data products and is derived from SPICE calculations. The attributes included in this class are as follows:

```
lucy:target_fov_count
lucy:target_fov_name
```

The `target_fov_count` attribute is the only required element in the class.

### 3.4 Lucy\_Product\_Information

The `Lucy_Product_Information` class and attributes describe any *Lucy* data processing pipeline specific data product information. Attributes collected in this class do not have counterparts in any PDS4 discipline dictionary. For general data processing information the processing dictionary (`proc:`) is used. The attributes found in the `Lucy_Product_Information` are as follows:

```
lucy:internal_product_version_id
```

The `internal_product_version_id` is the only element in this class and is required. It is possible that additional attributes could be added over the course of the mission.

### 3.5 LLORRI\_Instrument\_Parameters

The LLORRI\_Instrument\_Parameters class is the wrapper class for all L'ORRI instrument specific parameters. This class only appears in L'ORRI data products. The LLORRI\_Instrument\_Parameters class contains the attributes and sub-classes listed below:

```
lucy:attached_sync_marker_dec
lucy:attached_sync_marker_hex
lucy:latch_count
lucy:calibration_lamp_level_1
lucy:calibration_lamp_level_2
lucy:bias_level
lucy:read_noise
lucy:ccd_gain
lucy:bias_offset
lucy:frame_transfer_scrub_time
Radiometric_Conversion_Constants
```

The lucy:Radiometric\_Conversion\_Constants class provides the values needed to perform the conversion from the calibrated pixel values to physical units. This conversion is dependent on the spectral distribution of the source, the detector wavelength range, and whether or not the source is resolved. A description of how to convert from data values to physical units is provided in the LLORRI SIS document. The sub-class contains attributes and sub-sub-classes as follows:

```
lucy:pivot_wavelength
lucy:Diffuse_Source
lucy:Point_Source
lucy:photometric_zero_point
```

All attributes and classes are required for the Radiometric\_Conversion\_Constants class.

The lucy:Diffuse\_Source sub-class presents radiometric conversion constants for converting calibrated pixel values into physical units for unresolved sources.

```
lucy:units_of_conversion_constants
lucy:solar_constant
lucy:red_trojan_constant
lucy:gray_trojan_constant
lucy:dinkinesh_constant
```

The units, solar, red and gray constant attributes are required. Target specific constants are not required and will be extended as the mission progresses.

The lucy:Point\_Source sub-class presents radiometric conversion constants for converting calibrated pixel values into physical units for resolved/extended sources.

```
lucy:units_of_conversion_constants
lucy:solar_constant
lucy:red_trojan_constant
lucy:gray_trojan_constant
lucy:dinkinesh_constant
```

The units, solar, red and gray constant attributes are required. Target specific constants are not required and will be extended as the mission progresses.

### 3.6 LRalph\_Instrument\_Common\_Parameters

The LRalph\_Instrument\_Common\_Parameters class is a set of instrument attributes that are specific to the L'Ralph instrument suite which comprises the LEISA and MVIC components. This class only appears in LEISA or MVIC data products. The LRalph\_Instrument\_Common\_Parameters class contains the following attributes:

```
lucy:playback_start
lucy:playback_end
lucy:playback_bad_pixel_mask_status
lucy:playback_truncation_status
lucy:playback_header_type
lucy:test_pattern_setting
lucy:hs_xtrack_start_row
lucy:scan_row_pixels
lucy:fpecrc_error_flag
lucy:m4_calibration_state
lucy:m4_blackbody_setting
lucy:m4_filament_setting
lucy:fpe_drop_frames
lucy:mce_scan_rate
lucy:first_packet_timestamp
lucy:last_packet_timestamp
lucy:xsum_error_flag_status
lucy:lr_acquisition_start_block
lucy:mce_start_postion
lucy:mce_end_postion
lucy:board_selection
lucy:blocks_allocated
```

None of the attributes are required for this class.

### 3.7 LEISA\_Instrument\_Parameters

The LEISA\_Instrument\_Parameters class contains attributes specific to the LEISA instrument and only appears in LEISA data products. The attributes contained in this class are as follows:

```
lucy:leisa_mode
lucy:leisa_observation_allocation
lucy:leisa_integration_time
lucy:leisa_atrack_start_channel
lucy:leisa_atrack_num_channels
lucy:leisa_xtrack_start_row
lucy:leisa_xtrack_num_rows
lucy:leisa_offset
lucy:leisa_fpa_temperature
```

None of the attributes are required.

### 3.8 MVIC\_Instrument\_Parameters

The MVIC\_Instrument\_Parameters class contains attributes specific to the MVIC instrument and only appears in LEISA dt products. The attributes contained in this class are as follows:

```
lucy:mVIC_ccd_list
lucy:mVIC_actual_atrack_summing
lucy:mVIC_actual_xtrack_summing
lucy:mVIC_planned_atrack_summing
lucy:mVIC_planned_xtrack_summing
lucy:mVIC_summing_mode
lucy:actual_ccd1_tdi_configuration
lucy:actual_ccd2_tdi_configuration
lucy:actual_ccd3_tdi_configuration
lucy:actual_ccd4_tdi_configuration
lucy:actual_ccd5_tdi_configuration
lucy:actual_ccd6_tdi_configuration
lucy:planned_ccd1_tdi_configuration
lucy:planned_ccd2_tdi_configuration
lucy:planned_ccd3_tdi_configuration
lucy:planned_ccd4_tdi_configuration
lucy:planned_ccd5_tdi_configuration
lucy:planned_ccd6_tdi_configuration
lucy:mVIC_fpe_vis_temperature
lucy:mVIC_observation_allocation
lucy:mVIC_tdi_row_integration_time
```

None of the attributes are required.





## DEFINITIONS

| Type      | Name                              | Description                                                                                              |
|-----------|-----------------------------------|----------------------------------------------------------------------------------------------------------|
| Class     | Lucy_Observation_Time_Information | Lucy Mission observation identification and timing information.                                          |
| Attribute | observation_id                    | The lucy:observation_id attribute appears in all mission science data products.                          |
| Attribute | observation_id_count              | The lucy:observation_id_count attribute appears in all mission science data products.                    |
| Attribute | observation_complete              | The lucy:observation_complete attribute appears in all mission science data products.                    |
| Attribute | observation_missing_packets       | The lucy:observation_missing_packets attribute appears in all mission science data products.             |
| Attribute | start_sclk                        | The lucy:start_sclk attribute appears in all mission science data products and reports.                  |
| Attribute | mid_sclk                          | The lucy:mid_sclk attribute appears in all mission science data products and reports.                    |
| Attribute | stop_sclk                         | The lucy:stop_sclk attribute appears in all mission science data products and reports.                   |
| Attribute | mid_utc                           | The lucy:mid_utc attribute appears in all mission science data products and reports.                     |
| Attribute | mid_sclk_string                   | The lucy:mid_sclk_string attribute appears in all mission science data products and reports.             |
| Attribute | mid_utc_doy                       | The lucy:mid_utc_doy attribute appears in all mission science data products and reports.                 |
| Attribute | mid_utc_jd                        | The lucy:mid_utc_jd attribute appears in all mission science data products and reports.                  |
| Attribute | mid_ephemeris_time                | The lucy:mid_ephemeris_time attribute appears in all mission science data products and reports.          |
| Attribute | ccsds_sclk_time                   | The lucy:ccsds_sclk_time attribute appears in all mission science data products and reports.             |
| Class     | Lucy_Observation_Planning         | The Lucy Mission observation planning class and attributes describe the observations.                    |
| Attribute | load_identifier                   | The lucy:load_identifier attribute appears in all mission science data products and reports.             |
| Attribute | mission_segment                   | The lucy:mission_segment attribute appears in all mission science data products and reports.             |
| Attribute | sap_identifier                    | The lucy:sap_identifier attribute appears in all mission science data products and reports.              |
| Attribute | visit_name                        | The lucy:visit_name attribute appears in all mission science data products and reports.                  |
| Attribute | instrument_side                   | The lucy:instrument_side attribute appears in all mission science data products and reports.             |
| Attribute | llorri_status                     | The lucy:llorri_status attribute appears in all mission science data products and reports.               |
| Attribute | lralph_status                     | The lucy:lralph_status attribute appears in all mission science data products and reports.               |
| Attribute | ltes_status                       | The lucy:ltes_status attribute appears in all mission science data products and reports.                 |
| Attribute | ttcam_status                      | The lucy:ttcam_status attribute appears in all mission science data products and reports.                |
| Class     | Lucy_Target_List                  | The Lucy Mission class that contains the target list of potential targets in a given observation.        |
| Attribute | target_fov_count                  | The lucy:target_fov_count attribute appears in all mission science data products and reports.            |
| Attribute | target_fov_name                   | The lucy:target_fov_name attribute appears in all mission science data products and reports.             |
| Class     | Lucy_Product_Information          | The Lucy Mission class that contains attributes specific to the labeled data products.                   |
| Attribute | internal_product_version_id       | The lucy:internal_product_version_id attribute appears in all mission science data products and reports. |
| Class     | LLORRI_Instrument_Parameters      | The Lucy LORRI instrument class that contains instrument specific attributes.                            |
| Attribute | attached_sync_marker_dec          | The lucy:attached_sync_marker_dec attribute appears in L'LORRI science data products and reports.        |
| Attribute | attached_sync_marker_hex          | The lucy:attached_sync_marker_hex attribute appears in L'LORRI science data products and reports.        |
| Attribute | latch_count                       | The lucy:latch_count attribute appears in L'LORRI science data products and reports.                     |
| Attribute | calibration_lamp_level_1          | The lucy:calibration_lamp_level_1 attribute appears in L'LORRI science data products and reports.        |
| Attribute | calibration_lamp_level_2          | The lucy:calibration_lamp_level_2 attribute appears in L'LORRI science data products and reports.        |
| Attribute | bias_level                        | The lucy:bias_level attribute appears in calibrated L'LORRI data products and reports.                   |
| Attribute | read_noise                        | The lucy:read_noise attribute appears in calibrated L'LORRI data products and reports.                   |
| Attribute | ccd_gain                          | The lucy:ccd_gain attribute appears in calibrated L'LORRI data products and reports.                     |

| Type      | Name                                | Description                                                                                             |
|-----------|-------------------------------------|---------------------------------------------------------------------------------------------------------|
| Attribute | bias_offset                         | The lucy:bias_offset attribute appears in calibrated L’LORRI data products and reports the              |
| Attribute | frame_transfer_scrub_time           | The lucy:frame_transfer_scrub_time attribute appears in calibrated L’LORRI data products and            |
| Class     | LRalph_Instrument_Common_Parameters | The Lucy LRalph instrument class that contains the common (LEISA and MVIC) science data                 |
| Attribute | playback_start                      | The lucy:playback_start attribute appears in L’Ralph (MVIC and LEISA) science data products             |
| Attribute | playback_end                        | The lucy:playback_end attribute appears in L’Ralph (MVIC and LEISA) science data products               |
| Attribute | playback_bad_pixel_mask_status      | The lucy:playback_bad_pixel_mask_status attribute appears in L’Ralph (MVIC and LEISA) science           |
| Attribute | playback_truncation_status          | The lucy:playback_truncation attribute appears in L’Ralph (MVIC and LEISA) science data products        |
| Attribute | playback_header_type                | The lucy:playback_header_type attribute appears in L’Ralph (MVIC and LEISA) science data products       |
| Attribute | test_pattern_setting                | The lucy:test_pattern_setting attribute appears in L’Ralph (MVIC and LEISA) science data products       |
| Attribute | hs_xtrack_start_row                 | The lucy:hs_xtrack_start_row attribute appears in L’Ralph (MVIC and LEISA) science data products        |
| Attribute | scan_row_pixels                     | The lucy:scan_row_pixels attribute appears in L’Ralph (MVIC and LEISA) science data products            |
| Attribute | scan_row_pixels                     | The lucy:scan_row_pixels attribute appears in L’Ralph (MVIC and LEISA) science data products            |
| Attribute | fpecrc_error_flag                   | The lucy:fpecrc_error_flag attribute appears in L’Ralph (MVIC and LEISA) science data products          |
| Attribute | m4_calibration_state                | The lucy:m4_calibration_state attribute appears in L’Ralph (MVIC and LEISA) science data products       |
| Attribute | m4_blackbody_setting                | The lucy:m4_blackbody_state attribute appears in L’Ralph (MVIC and LEISA) science data products         |
| Attribute | m4_filament_setting                 | The lucy:m4_filament_state attribute appears in L’Ralph (MVIC and LEISA) science data products          |
| Attribute | fpe_drop_frames                     | The lucy:fpe_drop_frames attribute appears in L’Ralph (MVIC and LEISA) science data products            |
| Attribute | mce_scan_rate                       | The lucy:mce_scan_rate attribute appears in L’Ralph (MVIC and LEISA) science data products              |
| Attribute | first_packet_timestamp              | The lucy:first_packet_timestamp appears in L’Ralph (MVIC and LEISA) science data products               |
| Attribute | last_packet_timestamp               | The lucy:last_packet_timestamp attribute appears in L’Ralph (MVIC and LEISA) science data products      |
| Attribute | xsum_error_flag_status              | The lucy:xsum_error_flag_status attribute appears in L’Ralph (MVIC and LEISA) science data products     |
| Attribute | lr_acquisition_start_block          | The lucy:lr_acquisition_start_block attribute appears in L’Ralph (MVIC and LEISA) science data products |
| Attribute | mce_start_postion                   | The lucy:mce_start_postion attribute appears in L’Ralph (MVIC and LEISA) science data products          |
| Attribute | mce_end_postion                     | The lucy:mce_end_postion attribute appears in L’Ralph (MVIC and LEISA) science data products            |
| Attribute | board_selection                     | The lucy:board_selection attribute appears in L’Ralph (MVIC and LEISA) science data products            |
| Attribute | blocks_allocated                    | The lucy:blocks_allocated attribute appears in L’Ralph (MVIC and LEISA) science data products           |
| Class     | LEISA_Instrument_Parameters         | The Lucy LRalph LEISA class that contains the instrument specific attributes and science data           |
| Attribute | leisa_mode                          | The lucy:leisa_mode attribute appears in LEISA science data products and reports the                    |
| Attribute | leisa_observation_allocation        | The lucy:leisa_observation_allocation attribute appears in LEISA science data products and              |
| Attribute | leisa_integration_time              | The lucy:leisa_integration_time attribute appears in LEISA science data products and reports the        |
| Attribute | leisa_atrack_start_channel          | The lucy:leisa_atrack_start_channel appears in LEISA science data products and reports the              |
| Attribute | leisa_atrack_num_channels           | The lucy:leisa_atrack_num_channels appears in LEISA science data products and reports the               |
| Attribute | leisa_xtrack_start_row              | The lucy:leisa_xtrack_start_row appears in LEISA science data products and reports the                  |
| Attribute | leisa_xtrack_num_rows               | The lucy:leisa_xtrack_num_row appears in LEISA science data products and reports the                    |
| Attribute | leisa_offset                        | The lucy:leisa_offset appears in LEISA science data products and reports the                            |
| Attribute | leisa_fpa_temperature               | The lucy:leisa_offset appears in LEISA science data products and reports the                            |
| Class     | MVIC_Instrument_Parameters          | The Lucy LRalph MVIC class that contains the instrument specific attributes and science data            |
| Attribute | mvic_ccd_list                       | The lucy:mvic_ccd_list attribute appears in MVIC science data products and reports the                  |
| Attribute | mvic_actual_atrack_summing          | The lucy:mvic_actual_atrack_summing attribute appears in MVIC science data products and                 |
| Attribute | mvic_actual_xtrack_summing          | The lucy:mvic_actual_xtrack_summing attribute appears in MVIC science data products and                 |
| Attribute | mvic_planned_atrack_summing         | The lucy:mvic_planned_atrack_summing attribute appears in MVIC science data products and                |
| Attribute | mvic_planned_xtrack_summing         | The lucy:mvic_planned_xtrack_summing attribute appears in MVIC science data products and                |
| Attribute | mvic_summing_mode                   | The lucy:mvic_summing_mode attribute appears in MVIC science data products and reports the              |
| Attribute | actual_ccd1_tdi_configuration       | The lucy:actual_ccd1_tdi_configuration attribute appears in MVIC science data products and              |
| Attribute | actual_ccd2_tdi_configuration       | The lucy:actual_ccd2_tdi_configuration attribute appears in MVIC science data products and              |
| Attribute | actual_ccd3_tdi_configuration       | The lucy:actual_ccd3_tdi_configuration attribute appears in MVIC science data products and              |
| Attribute | actual_ccd4_tdi_configuration       | The lucy:actual_ccd4_tdi_configuration attribute appears in MVIC science data products and              |
| Attribute | actual_ccd5_tdi_configuration       | The lucy:actual_ccd5_tdi_configuration attribute appears in MVIC science data products and              |
| Attribute | actual_ccd6_tdi_configuration       | The lucy:actual_ccd6_tdi_configuration attribute appears in MVIC science data products and              |
| Attribute | planned_ccd1_tdi_configuration      | The lucy:planned_ccd1_tdi_configuration attribute appears in MVIC science data products and             |

| Type      | Name                             | Description                                                                    |
|-----------|----------------------------------|--------------------------------------------------------------------------------|
| Attribute | planned_ccd2_tdi_configuration   | The lucy:planned_ccd2_tdi_configuration attribute appears in MVIC science data |
| Attribute | planned_ccd3_tdi_configuration   | The lucy:planned_ccd3_tdi_configuration attribute appears in MVIC science data |
| Attribute | planned_ccd4_tdi_configuration   | The lucy:planned_ccd4_tdi_configuration attribute appears in MVIC science data |
| Attribute | planned_ccd5_tdi_configuration   | The lucy:planned_ccd5_tdi_configuration attribute appears in MVIC science data |
| Attribute | planned_ccd6_tdi_configuration   | The lucy:planned_ccd6_tdi_configuration attribute appears in MVIC science data |
| Attribute | mvic_fpe_vis_temperature         | The lucy:mvic_fpe_vis_temperature attribute appears in MVIC science data       |
| Attribute | mvic_observation_allocation      | The lucy:mvic_observation_allocation attribute appears in MVIC science data    |
| Attribute | mvic_tdi_row_integration_time    | The lucy:mvic_tdi_row_integration_time attribute appears in MVIC science data  |
| Class     | Point_Source                     | The lucy:Point_Source class presents radiometric conversion constants for co   |
| Attribute | units_of_conversion_constants    | For resolved sources, the radiometric conversion produces data in units of ra  |
| Attribute | solar_constant                   | The lucy:solar_constant attribute supplies the constant to be used for sources |
| Attribute | red_trojan_constant              | The lucy:red_trojan_constant attribute supplies the constant to be used for so |
| Attribute | gray_trojan_constant             | The lucy:gray_trojan_constant attribute supplies the constant to be used for s |
| Attribute | dinkinesh_constant               | The lucy:dinkinesh_constant attribute supplies the constant to be used for so  |
| Class     | Radiometric_Conversion_Constants | The lucy:Radiometric_Conversion_Constants class provides the values need       |
| Attribute | pivot_wavelength                 | The value contained in the lucy:pivot_wavelength attribute is used when con    |
| Attribute | photometric_zero_point           | The lucy:photometric_zero_point attribute supplies the photometric zero poi    |
| Class     | Diffuse_Source                   | The lucy:Point_Source class presents radiometric conversion constants for co   |
| Attribute | units_of_conversion_constants    | For resolved sources, the radiometric conversion produces data in units of ra  |
| Attribute | solar_constant                   | The lucy:solar_constant attribute supplies the constant to be used for sources |
| Attribute | red_trojan_constant              | The lucy:red_trojan_constant attribute supplies the constant to be used for so |
| Attribute | gray_trojan_constant             | The lucy:gray_trojan_constant attribute supplies the constant to be used for s |
| Attribute | dinkinesh_constant               | The lucy:dinkinesh_constant attribute supplies the constant to be used for so  |



## EXAMPLES

Note: the examples below reside within the Mission\_Area section of their PDS4 labels.

## 5.1 Observational parameters common to all instruments

```
<lucy:Lucy_Observation_Planning>
  <lucy:load_identifier>22289 </lucy:load_identifier>
  <lucy:mission_segment>EGA_1</lucy:mission_segment>
  <lucy:sap_identifier>SAP.RA.C.013.v01__Earth_Radiometry</lucy:sap_identifier>
  <lucy:visit_name>22289.ega1.RA.C.013.v01.288ae__Earth_Radiometry</lucy:visit_name>
  <lucy:instrument_side>SIDE_1</lucy:instrument_side>
  <lucy:llorri_status>UNKNOWN</lucy:llorri_status>
  <lucy:lralth_status>ACTIVE</lucy:lralth_status>
  <lucy:ltes_status>ACTIVE</lucy:ltes_status>
  <lucy:ttcam_status>UNKNOWN</lucy:ttcam_status>
</lucy:Lucy_Observation_Planning>
<lucy:Lucy_Observation_Time_Information>
  <lucy:observation_id>2230</lucy:observation_id>
  <lucy:observation_complete>COMPLETE</lucy:observation_complete>
  <lucy:observation_missing_packets>0</lucy:observation_missing_packets>
  <lucy:start_sclk unit="s"> 719078053.248</lucy:start_sclk>
  <lucy:mid_sclk unit="s">719078082.0999761 </lucy:mid_sclk>
  <lucy:stop_sclk unit="s">719078110.9519521</lucy:stop_sclk>
  <lucy:mid_utc>2022-10-15T03:54:59.454Z</lucy:mid_utc>
  <lucy:mid_sclk_string>1/0719078082-06552</lucy:mid_sclk_string>
  <lucy:mid_utc_doy>2022-288T03:54:59.454Z</lucy:mid_utc_doy>
  <lucy:mid_utc_jd unit="julian day">JD 2459867.6631881</lucy:mid_utc_jd>
  <lucy:mid_ephemeris_time unit="s">719078168.6364158 </lucy:mid_ephemeris_time>
  <lucy:ccsds_sclk_time unit="s"> 71942620</lucy:ccsds_sclk_time>
</lucy:Lucy_Observation_Time_Information>
<lucy:Lucy_Target_List>
  <lucy:target_fov_count>1</lucy:target_fov_count>
  <lucy:target_fov_name>Earth</lucy:target_fov_name>
</lucy:Lucy_Target_List>
<lucy:Lucy_Product_Information>
  <lucy:internal_product_version_id>1</lucy:internal_product_version_id>
</lucy:Lucy_Product_Information>
```

## 5.2 L'LORRI instrument parameters

```

<lucy:LLORRI_Instrument_Parameters>
    <lucy:attached_sync_marker_dec>449838109</lucy:attached_sync_marker_dec>
    <lucy:attached_sync_marker_hex>0x1ACFFC1D</lucy:attached_sync_marker_hex>
    <lucy:latch_count>0</lucy:latch_count>
    <lucy:calibration_lamp_level_1 unit="DN">0</lucy:calibration_lamp_level_
→ 1>
    <lucy:calibration_lamp_level_2 unit="DN">0</lucy:calibration_lamp_level_
→ 2>
    <lucy:bias_level unit="DN">120.0531862745098</lucy:bias_level>
    <lucy:read_noise unit="DN">0.9</lucy:read_noise>
    <lucy:ccd_gain unit="electron/DN">21.1</lucy:ccd_gain>
    <lucy:bias_offset unit="DN">3.2</lucy:bias_offset>
    <lucy:frame_transfer_scrub_time unit="ms">11.7762</lucy:frame_transfer_
→ scrub_time>
    <lucy:Radiometric_Conversion_Constants>
        <lucy:pivot_wavelength unit="Angstrom">6030.0</lucy:pivot_wavelength>
        <lucy:Diffuse_Source>
            <lucy:units_of_conversion_constants>(DN/s/pixel)/(erg/cm^2/s/
→ Angstrom/sr)</lucy:units_of_conversion_constants>
            <lucy:solar_constant>271300.0</lucy:solar_constant>
            <lucy:red_trojan_constant>278400.0</lucy:red_trojan_constant>
            <lucy:gray_trojan_constant>271200.0</lucy:gray_trojan_constant>
            <lucy:dinkinesh_constant>286000.0</lucy:dinkinesh_constant>
        </lucy:Diffuse_Source>
        <lucy:Point_Source>
            <lucy:units_of_conversion_constants>(DN/s)/(erg/cm^2/s/Angstrom)
→ </lucy:units_of_conversion_constants>
            <lucy:solar_constant>1.101e+16</lucy:solar_constant>
            <lucy:red_trojan_constant>1.13e+16</lucy:red_trojan_constant>
            <lucy:gray_trojan_constant>1.101e+16</lucy:gray_trojan_constant>
        </lucy:Point_Source>
        <lucy:photometric_zero_point>18.93</lucy:photometric_zero_point>
    </lucy:Radiometric_Conversion_Constants>
</lucy:LLORRI_Instrument_Parameters>

```

## 5.3 L'Ralph common instrument parameters (shared between MVIC and LEISA)

```

<lucy:LRalph_Instrument_Common_Parameters>
    <lucy:playback_start unit="s">719548332</lucy:playback_start>
    <lucy:playback_end unit="s">719549015</lucy:playback_end>
    <lucy:playback_bad_pixel_mask_status>>false</lucy:playback_bad_pixel_mask_status>
    <lucy:playback_truncation_status>true</lucy:playback_truncation_status>
    <lucy:playback_header_type>MVIC</lucy:playback_header_type>
    <lucy:test_pattern_setting>0</lucy:test_pattern_setting>
    <lucy:hs_xtrack_start_row>79</lucy:hs_xtrack_start_row>
    <lucy:scan_row_pixels>5024</lucy:scan_row_pixels>
    <lucy:fpecrc_error_flag>>false</lucy:fpecrc_error_flag>

```

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```

<lucy:m4_calibration_state>0</lucy:m4_calibration_state>
<lucy:m4_blackbody_setting>0</lucy:m4_blackbody_setting>
<lucy:m4_filament_setting>0</lucy:m4_filament_setting>
<lucy:fpe_drop_frames>25</lucy:fpe_drop_frames>
<lucy:mce_scan_rate unit="rad/s">0.004</lucy:mce_scan_rate>
<lucy:first_packet_timestamp unit="s">719212908.1950001 </lucy:first_packet_timestamp>
<lucy:last_packet_timestamp unit="s">719212919.048</lucy:last_packet_timestamp>
<lucy:xsum_error_flag_status>>false</lucy:xsum_error_flag_status>
<lucy:lr_acquisition_start_block>2419</lucy:lr_acquisition_start_block>
<lucy:mce_start_postion>-18200</lucy:mce_start_postion>
<lucy:mce_end_postion>17000</lucy:mce_end_postion>
<lucy:board_selection>FM 1</lucy:board_selection>
<lucy:blocks_allocated>24</lucy:blocks_allocated>
</lucy:L'Ralp Instrument Common Parameters>

```

## 5.4 L'Ralp/MVIC-specific instrument parameters

```

<lucy:MVIC_Instrument_Parameters>
  <lucy:mVIC_ccd_list>2,3,4,5,6</lucy:mVIC_ccd_list>
  <lucy:mVIC_actual_atrack_summing>0</lucy:mVIC_actual_atrack_summing>
  <lucy:mVIC_actual_xtrack_summing>0</lucy:mVIC_actual_xtrack_summing>
  <lucy:mVIC_planned_atrack_summing>0</lucy:mVIC_planned_atrack_summing>
  <lucy:mVIC_planned_xtrack_summing>0</lucy:mVIC_planned_xtrack_summing>
  <lucy:mVIC_summing_mode>0</lucy:mVIC_summing_mode>
  <lucy:actual_ccd1_tdi_configuration>NO_PLAYBACK</lucy:actual_ccd1_tdi_configuration>
  <lucy:actual_ccd2_tdi_configuration>TDI_4 </lucy:actual_ccd2_tdi_configuration>
  <lucy:actual_ccd3_tdi_configuration>TDI_4 </lucy:actual_ccd3_tdi_configuration>
  <lucy:actual_ccd4_tdi_configuration>TDI_4 </lucy:actual_ccd4_tdi_configuration>
  <lucy:actual_ccd5_tdi_configuration>TDI_4 </lucy:actual_ccd5_tdi_configuration>
  <lucy:actual_ccd6_tdi_configuration>TDI_4 </lucy:actual_ccd6_tdi_configuration>
  <lucy:planned_ccd1_tdi_configuration>NO_PLAYBACK</lucy:planned_ccd1_tdi_configuration>
  <lucy:planned_ccd2_tdi_configuration>4_ROWS </lucy:planned_ccd2_tdi_configuration>
  <lucy:planned_ccd3_tdi_configuration>4_ROWS </lucy:planned_ccd3_tdi_configuration>
  <lucy:planned_ccd4_tdi_configuration>4_ROWS </lucy:planned_ccd4_tdi_configuration>
  <lucy:planned_ccd5_tdi_configuration>4_ROWS </lucy:planned_ccd5_tdi_configuration>
  <lucy:planned_ccd6_tdi_configuration>4_ROWS </lucy:planned_ccd6_tdi_configuration>
  <lucy:mVIC_fpe_vis_temperature unit ="degC">9.009289875250857</lucy:mVIC_fpe_vis_
  →temperature>
  <lucy:mVIC_observation_allocation unit ="s">10</lucy:mVIC_observation_allocation>
  <lucy:mVIC_tdi_row_integration_time unit ="microseconds">7250</lucy:mVIC_tdi_row_
  →integration_time>
</lucy:MVIC_Instrument_Parameters>

```

## 5.5 L'Ralph/LEISA-specific instrument parameters

```
<lucy:LEISA_Instrument_Parameters>
  <lucy:leisa_mode>T</lucy:leisa_mode>
  <lucy:leisa_observation_allocation unit="s">59</lucy:leisa_observation_allocation>
  <lucy:leisa_integration_time unit="microseconds">26.640</lucy:leisa_integration_time>
  <lucy:leisa_atrack_start_channel>4</lucy:leisa_atrack_start_channel>
  <lucy:leisa_atrack_num_channels>23</lucy:leisa_atrack_num_channels>
  <lucy:leisa_xtrack_start_row>647</lucy:leisa_xtrack_start_row>
  <lucy:leisa_xtrack_num_rows>20</lucy:leisa_xtrack_num_rows>
  <lucy:leisa_offset>1111</lucy:leisa_offset>
  <lucy:leisa_fpe_ir_temperature unit="degC">-1.701436782829319</lucy:leisa_fpe_ir_
  ↪temperature>
</lucy:LEISA_Instrument_Parameters>
```



***LUCY* MISSION LDD**

Steward: M. Katherine Combie

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Detailed documentation is in development. In the meantime, refer to the user's guide sections for information about this dictionary.